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Cover member and thermostat device equipped with cover member

Abstract

To prevent a gasket fitted into a groove of a housing of a thermostat device from coming off and the leakage of coolant. A cover member detachably attached to a thermostat device includes a housing that has a body having an opening at one end and a groove surrounding the opening and a pair of legs standing inside the groove from one end of the body, and a gasket to be fitted into the groove. The cover member has a dish-shaped portion that has an insertion hole through which the pair of legs is inserted and covers the gasket, and a cylindrical portion erecting from the opening edge of the insertion hole and into which the pair of legs are press-fitted inside, and a corrugated section is formed in the portion of the cylindrical portion opposite to the legs.

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Background/Summary

RELATED APPLICATIONS

(1) The present application is National Phase of International Application No. PCT/JP2022/016523 filed Mar. 31, 2022, and claims priority from Japanese Application No. 2021-099122, filed Jun. 15, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) This invention relates to a cover member of a gasket attachable to a thermostat device and a thermostat device with the cover member.

BACKGROUND ART

(3) A thermostat device is equipped in a cooling circuit of an engine (an internal combustion engine) and adjusts the temperature of the coolant. The housing of the thermostat device has an opening on one end, and the edge of the opening contacts closely to a counterpart member (a water pump and a water jacket of the engine, for example) through a gasket as a sealing member. This prevents the leakage of the coolant in the housing.

(4) More specifically, an annular groove surrounding the opening is formed at the opening end of the housing (Patent Literature 1, for example), and an annular gasket is fitted to the groove (Patent Literature 2, for example).

CITATION LIST

Patent Literature

(5) PTL 1: JP-A-2020-169611 PTL 2: Japanese Patent No. 3310547

SUMMARY OF INVENTION

Technical Problem

(6) The gasket recited in PTL 2 has projecting parts on the inner and outer circumferences to prevent it from coming off. This prevents the gasket from coming off the groove due to vibration when transporting the thermostat device to the counterpart member before assembly. However, to keep the gland fill of the gasket to an appropriate value, the number of projecting parts to prevent the coming-off cannot be increased without limits; the provision of coming-off-preventing projecting parts alone may not be sufficient to reliably prevent the gasket from coming off the groove.

(7) The present invention is made in considering the above problem. It aims to provide a cover member that reliably prevents the gasket from coming off and a thermostat device with the cover member.

Solution to Problem

(8) As for a first means to solve the above problem, a cover member according to the present

invention is the one that is detachably attached to the thermostat device. The thermostat device is provided with a housing that includes a body having an opening at one end and a groove formed to surround the opening and a pair of legs standing up opposingly from one end of the body inside the groove, and also with a gasket to be fitted into the groove. The cover member is provided with a dish-shaped portion covering the gasket having insertion holes inside in which the pair of legs are inserted and a cylindrical portion standing up from the opening edge of the insertion holes, inside of which the pair of legs are press-fitted, and a corrugated section is formed at a portion opposing to the legs of the cylindrical portion.

(9) With this, in a state where the cover member is attached to the thermostat device, since the gasket is covered with the dish-shaped portion, the gasket will not come off.

(10) As for a second means to solve the above problem, a thermostat device, according to the present invention, with a cover member is provided with a thermostat device and a cover member to be detachably attached to the thermostat device.

(11) The thermostat device includes a housing being provided with a body that has an opening at one end and a groove formed so as to surround the opening and a pair of legs standing up opposing to each other at the one end of the body inside the groove, and a gasket to be fitted in the groove. The cover member additionally includes a dish-shaped portion that has insertion holes inside, through which the pair of legs are to be inserted, and covers the gasket, and a cylindrical portion standing up from the opening edge of the insertion holes and to which the pair of legs is to be press-fitted thereinside, and a corrugated section is formed at the portion of the cylindrical portion opposing to the legs.

(12) With this, in a state where the cover member is attached to the thermostat device, since the gasket is covered with the dish-shaped portion, the gasket will not come off.

(13) Further, the thermostat device includes a valve seat formed inside the body, a thermo-element, one end of which is supported by the body, extending and retracting depending on the temperature, a valve body unseated from or seated on the valve seat in response to the extension-retraction operation of the thermo-element, a biasing member biasing the valve body to the valve seat side, and a frame being supported by the pair of legs to support one end of the biasing member. An inward-convex portion protruding inward is formed inside the cylindrical portion of the cover member; the inward-convex portion is located at the body side from the frame in a state where the cover is attached to the thermostat device. When the cover member is attached to or detached from the thermostat device, the inward-convex portion may be made in contact with the frame to deform elastically.

(14) This allows to prevent the cover member from coming off from the thermostat device unintentionally due to vibration during transportation.

(15) Further, as for a third means to solve the above problem, the cover member, according to the present invention, is a cover detachably attached to the thermostat device. The thermostat device includes a housing being provided with a body that has an opening at one end and a groove formed so as to surround the opening and a pair of legs standing up opposing to each other at the one end of the body inside the groove, a gasket to be fitted in the groove, a valve seat formed inside the body, a thermo-element, one end of which is supported by the body, extending and retracting depending on the temperature, a valve body unseated from or seated on the valve seat in response to the extension-retraction operation of the thermo-element, a biasing member biasing the valve body to the valve seat side, and a frame being supported by the pair of legs to support one end of the biasing member. The cover member includes a dish-shaped portion that has insertion holes inside, through which the pair of legs are to be inserted, and covers the gasket, and a cylindrical portion standing up from the opening edge of the insertion holes and to which the pair of legs is to be inserted thereinside.

(16) An inward-convex portion protruding inward is formed on the cylindrical portion. The inward-convex portion is located at the body side from the frame in a state where the cover is attached to

the thermostat device. When the cover member is attached to or detached from the thermostat device, the inward-convex portion comes into contact with the frame to deform elastically.

(17) With this, in a state where the cover member is attached to the thermostat device, since the gasket is covered with the dish-shaped portion, the gasket will not come off.

(18) Further, as for a fourth means to solve the above problem, in the thermostat device with a cover member according to the present invention, the thermostat device includes a valve seat formed inside the body, a thermo-element, one end of which is supported by the body, extending and retracting depending on the temperature, a valve body unseated from or seated on the valve seat in response to the extension-retraction operation of the thermo-element, a biasing member biasing the valve body to the valve seat side, and a frame being supported by the pair of legs to support one end of the biasing member. An inward-convex portion protruding inward is formed on the cylindrical portion. The inward-convex portion is located at the body side from the frame in a state where the cover is attached to the thermostat device. When the cover member is attached to or detached from the thermostat device, the inward-convex portion comes into contact with the frame to deform elastically.

(19) With this, in a state where the cover member is attached to the thermostat device, since the gasket is covered with the dish-shaped portion, the gasket will not come off.

(20) For the cover member, one end of the cylindrical member may be closed; this prevents dust from entering the inside of the cover member, for example.

(21) For the cover member, the surface shape of the inward-convex portion may be spherical. With this, the cover member becomes easy to attach to the thermostat device. Then, the hooking margin of the inward-convex portion to the frame can be easily ensured, and the coming off of the cover member from the thermostat device due to the vibration during transportation can surely be suppressed.

(22) For the cover member, portions of the cylindrical portion located at the outside of the inward-convex portion may be recessed. This allows the inward-convex portion to elastically deform easily and enables the cover member to be mounted to the thermostat device. Then, the hooking margin of the inward-convex portion to the frame can be easily ensured, and the coming off of the cover member from the thermostat device due to the vibration during transportation can be further suppressed.

(23) Protrusions protruding outward of the dish-shaped portion may be formed on the gasket in the thermostat device to which the cover member is to be attached. With this, the gasket being attached can be visually confirmed easily, even in the case of the thermostat device with the cover member.

Advantageous Effects of the Invention

(24) With the cover member and the thermostat device with the cover member according to the present invention, the coming off of the gasket can be surely inhibited in the state where the cover member is attached to the thermostat device.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a side view of a thermostat device with the cover member of an embodiment to which a cover member is attached.
- (2) FIG. 2 is a bottom view of the thermostat device with the cover member of FIG. 1.
- (3) FIG. 3 is a cross-sectional view on arrow A-A of the thermostat device with the cover member of FIG. 2.
- (4) FIG. 4 is an enlarged cross-sectional view of part of the thermostat device in FIG. 3.
- (5) FIG. 5 is a cross-sectional view on arrow B-B of the thermostat device with the cover member of FIG. 2.

(6) FIG. 6 is a cross-sectional view on arrow C-C of the thermostat device with the cover member of FIG. 1.

(7) FIG. 7 is a cross-sectional view illustrating an enlarged part of FIG. 6.

DESCRIPTION OF EMBODIMENTS

(8) A cover member and a thermostat device with the cover member according to an embodiment of the present invention will be described based on the drawings below.

(9) FIG. 1 is a side view of a thermostat device with the cover member of an embodiment to which a cover member is attached, and FIG. 2 is a bottom view of the thermostat device with the cover member of FIG. 1. FIG. 3 is a cross-sectional view on arrow A-A of the thermostat device with the cover member of FIG. 2, and FIG. 4 is an enlarged cross-sectional view of part of the thermostat device in FIG. 3. FIG. 5 is a cross-sectional view on arrow B-B of the thermostat device with the cover member of FIG. 2. FIG. 6 is a cross-sectional view on arrow C-C of the thermostat device with the cover member of FIG. 1, and FIG. 7 is a cross-sectional view illustrating an enlarged part of FIG. 6.

(10) As shown in FIG. 1, the thermostat device with the cover member **100** includes a thermostat device **10** and a cover member **1** attached to the thermostat device **10**.

(11) The thermostat device **10** is mounted, with the cover member **1** detached, on a cooling circuit of a power engine including, for example, an engine (internal combustion engine), a battery, and a stack for fuel cells and controls the temperature of the coolant. Meanwhile, the cover member **1** is temporarily attached to the thermostat device **10** in a step before the assembly of the thermostat device **10** to a counterpart member (such as a water pump, a water jacket of the power engine, members for attaching the thermostat device **10**, or members constituting a cooling circuit) and is detached from the thermostat device **10** when the thermostat device **10** is assembled to the counterpart member.

(12) Here, a thermostat device to which a cover member is attached is called “a thermostat device with a cover member,” and a thermostat device from which the cover member is detached is simply called “a thermostat device.”

(13) First, the thermostat device **10** will be described. As described above, the thermostat device **10** is assembled to the counterpart member with the cover member **1** detached.

(14) The thermostat device **10** is, as shown in FIG. 3, provided with a housing **2**, a thermo-element **17** whose one end is supported by the housing **2** extending and retracting in response to the temperature, a valve body **15** that leaves from and is seated on a valve seat **20b**, to be described later, in the housing **2** in response to the extending and retracting operation of the thermo-element **17**, a coil spring **16** as a biasing member that biases the valve body **15** to the valve seat **20b** side, a frame **19** supporting one end of the coil spring **16**, and a guide member **26** hooked to the frame **19** guiding a temperature-sensitive case **30**, to be described later, of the thermo-element **17**.

(15) For the convenience of the description below, the upper and lower portions of the thermostat device **10** shown in FIG. 3 are simply referred to as “the upper” and “the lower.”

(16) In the present embodiment, the housing **2** is provided with a vertexed cylindrical body **20** at whose lower edge (one end) an opening **20c** is formed, a pair of flanges **3**, **3** protruding outward from the lower-end outer periphery of the body **20**, a pair of legs **21**, **21** extending downward opposing to each other from the lower end opening edge of the body **20** (standing up opposing to each other from the lower end opening edge of the body **20**), and a connecting port **23** of the radiator side extending obliquely upward from the top part of the body **20**.

(17) In the state where the thermostat device **10** has been assembled to the counterpart member, the coolant is configured to pass through the connecting port **23**, the inside of the body **20**, and the opening **20c**; the flow passage **4** in the housing **2** is composed of those elements.

(18) As shown in FIG. 2, one or two bolt holes **3a** are provided each on the pair of flanges **3**, **3**. Metal cylinders are press-fitted into the bolt holes **3a**, and bolts (not shown) for fastening the thermostat device **10** to the counterpart member are inserted in the cylinders. As shown in FIG. 3,

an annular groove **20d** is formed to surround the opening **20c**, at the lower end opening edge of the body **20** located inside the bolt holes **3a** (flange **3**), and a gasket **25** is mounted in the groove **20d**.

(19) The gasket **25** seals between the thermostat device **10** and the counterpart member and prevents the coolant flowing in the housing **2** from leaking outside in a state where the thermostat device **10** is attached to the counterpart member. The inside of the housing **2** is defined by the inside (inner side) of the gasket **25** at the body **20**.

(20) An annular valve seat **20b** is formed at the inner periphery of the lower end opening edge of the body **20** located inside the housing **2**; the flow passage **4** is opened or closed by the valve body **15** being unseated from or seated on the valve seat **20b**.

(21) The thermo-element **17** is inserted into the inside of the housing **2**. The thermo-element **17** is disposed such that the thermo-element **17** follows along the axial line of the axial part of the body **20**. The thermo-element **17** is provided with a temperature-sensitive case **30** incorporating a temperature-sensitive material such as wax changing the volume thereof in response to the temperature and a piston **18**, whose upper end is supported by the housing **2**, entering and exiting from the temperature-sensitive case **30** in response to the volume change of the temperature-sensitive material.

(22) When the temperature of the coolant surrounding the temperature-sensitive case **30** rises, the temperature-sensitive material therein expands, the piston **18** exits from the temperature-sensitive case **30**, and the thermo-element **17** extends. In contrast, when the temperature of the coolant surrounding the temperature-sensitive case **30** drops, the temperature-sensitive material therein contracts, the piston **18** enters the temperature-sensitive case **30**, and the thermo-element **17** retracts. Thus, the thermo-element **17** operates to extend or retract depending on the temperature.

(23) The tip end of the piston **18** located at the upper end of the thermo-element **17** is fitted to a cylinder-shaped boss **20a** formed inside at the top portion of the body **20**. Thus, the upward motion of the piston **18** against the housing **2** is inhibited. Therefore, when the thermo-element **17** operates to extend or retract, the temperature-sensitive case **30** moves down or up without changing the position of the piston **18** against the housing **2**.

(24) The valve body **15** is fixed to the outer circumferential surface of the temperature-sensitive case **30**. With this, the valve body **15** moves down or up with the temperature-sensitive case **30** associated with the extension or retraction of the thermo-element **17**. When the thermo-element **17** extends and the valve body **15** moves downward, the communication of the flow passage **4** is allowed because the valve body **15** leaves from the valve seat **20b**, and the coolant can pass through therebetween. In contrast, when the thermo-element **17** retracts and the valve body **15** moves upward to be seated on the valve seat **20b**, the communication of the flow passage **4** is shut down. Thus, the valve body **15** opens and closes the flow passage **4** by being unseated from and seated on the valve seat **20b**.

(25) The upper end of the coil spring **16** is in contact with the backside of the valve body **15**. The coil spring **16** is disposed so as to surround the thermo-element **17**. The lower end of the coil spring **16** is supported by the frame **19**.

(26) The periphery of the frame **19** is hooked to the tip end of the pair of legs **21**, **21** formed on the housing **2**, and the frame **19** is inhibited from moving downward against the housing **2**. A through-hole is formed at the center of the frame **19**. A guide member **26**, having a cylinder shape, is inserted into the through-hole, and the temperature-sensitive case **30** is inserted movably up and down inside the guide member **26**. That is, the temperature-sensitive case **30** is movable up and down against the frame **19** while guided by the guide member **26**.

(27) The guide member **26** includes a guiding portion **26a** having a cylinder shape inside which the temperature-sensitive case **30** is inserted and a brim **26b** protruding toward the outer periphery from the top end of the guiding portion **26a**. The brim **26b** is hooked to the frame **19** and put between the frame **19** and the coil spring **16**. With this, the guide member **26** is fixed to the frame

19. The guiding portion **26a** is cylindrical and has openings at the top and bottom, and additionally, multiple lateral holes **26al** penetrating the wall thickness are formed in the guiding portion **26a**. Because of this, the coolant can freely move inward and outward through the guiding portion **26a**.

(28) The coil spring **16** is a compressive spring and is disposed between the valve body **15** and the frame **19** in a compressed state. Thus, the valve body **15** is biased upward (toward the valve seat **20b** side) by the coil spring **16**. With this configuration, when the temperature of the coolant around the thermo-element **17** rises and the thermo-element **17** extends, the valve body **15** moves downward resisting the biasing force due to the coil spring **16** and leaves from the valve seat **20b**. Meanwhile, when the temperature of the coolant around the thermo-element **17** drops and the thermo-element **17** retracts, the valve body **15** moves upward by the biasing force due to the coil spring **16** and comes closer to the valve seat **20b**.

(29) Followingly, the cover member **1** according to the present embodiment will be described in detail. As described above, the cover member **1** is mounted to the thermostat device **10** in a step before the assembly of the thermostat device **10** to a counterpart member.

(30) More specifically, the cover member **1** is temporarily mounted on the thermostat device **10** to constitute a thermostat device with the cover member **100** with the thermostat device **10**, between the completion of the manufacturing process where the thermostat device **10** is manufactured as a part and the entering of the assembly process where the thermostat device **10** is attached to the counterpart member. The period between the manufacturing process and the assembly process includes a storing process where the thermostat device **10** is stored and a transportation process where the thermostat device **10** is moved and transported.

(31) The cover member **1** is formed of polypropylene (PP), for example, is transparent (including semitransparency), and has an almost uniform thickness all over the whole. In the present embodiment, the cover member **1** includes a dish-shaped portion **11** and a cylindrical portion **12** continuous therewith integrally.

(32) As shown in FIG. 3, the dish-shaped portion **11** has an insertion hole **11A** at its center. The pair of legs **21**, **21** is inserted in the insertion hole **11A** in a state where the cover member **1** is mounted on the thermostat device **10**. As shown in FIG. 2, the dish-shaped portion **11** is disposed to cover the gasket **25** in a state where the cover member **1** is mounted on the thermostat device **10**.

(33) In the present embodiment, the gasket **25** includes an annular sealing portion **25c** disposed to surround the opening **20c** of the housing **2** and multiple protrusions **25a**, **25b** protruding outward from the outer periphery of the sealing portion **25c**. In the gasket **25**, the sealing portion **25c** substantially functions as a sealing member to seal between the housing **2** and the counterpart member, and the dish-shaped portion **11** is formed to cover the entire sealing portion **25c**.

(34) The protrusions **25a** and **25b** have a length set to go outside from the dish-shaped portion **11**. In the present embodiment, the one protrusion **25a** has a length set to protrude to the further outer periphery side from the housing **2**. This enables the visual confirmation of the protrusion **25a** easily due to the conspicuousness of the protrusion **25a**, even in the case of the thermostat device with the cover member **100** where the cover member **1** is attached to the thermostat device **10**.

(35) The cylindrical portion **12** extends downward from an opening edge **11B** of the insertion hole **11A** at the center of the dish-shaped portion **11** (standing up from the opening edge **11B** of the insertion hole **11A**). As shown in FIG. 3, in the present embodiment, the cylindrical portion **12** is a bottomed cylinder, and the lower end is closed. Further, the pair of legs **21**, **21**, and the guide member **26** are inserted inside the cylindrical portion **12** in the state where the cover member **1** is mounted on the thermostat device **10**.

(36) The lower part of the cylindrical portion **12** has a small-diameter portion **12a** whose inner and outer diameters are smaller than any other part thereof. The guiding portion **26a** of the guide member **26** is inserted inside the reduce-diameter portion **12a**. The inner diameter of the small-diameter portion **12a** is greater than the outer diameter of the guiding portion **26a**; the guiding portion **26a** can be inserted inside the small-diameter portion **12a** without resistance.

(37) In the cylindrical portion **12**, portions opposing to the pair of legs **21, 21** of the thermostat device **10** with the cover member **1** are corrugated sections **13**. That is, a pair of corrugated sections **13, 13** are provided corresponding to the pair of legs **21, 21** in the cylindrical portion **12**. Let portions connected to the pair of corrugated sections **13, 13** along the circumferential direction be intermediate portions **12b**, respectively, as shown in FIGS. **1** and **2**, and the pair of the corrugated sections **13, 13** are disposed at the outer side of the intermediate portions **12b** (at the side leaving from the center of the cover member **1**) and are elastically deformable.

(38) The pair of the corrugated sections **13, 13** have a tightening margin against the pair of the legs **21, 21**. With this, at the time when the cover member **1** is attached to the thermostat device **10**, the pair of legs **21, 21** are inserted into the cylindrical portion **12** with elastically deforming the corrugated sections **13**. Thus, the cover member **1** is supported by the thermostat device **10** since the pair of the legs **21, 21** are press-fitted between the corrugated sections **13, 13** of the cylinder portion **12**.

(39) As shown in FIG. **4**, the corrugated section **13** has a form of mountain bending portions and valley bending portions which are alternately connected up and down, and the bending portions are curved, respectively.

(40) As shown in FIGS. **5** through **7**, in the inner circumferential surface of the cylindrical portion **12**, inward-convex portions **14** are provided at the positions upper side than the frame **19**, in the state where the cover member **1** is attached to the thermostat device **10**. In the present embodiment, the inward-convex portions **14** four in number are provided in the circumferential direction lined up at the intermediate portions **12b**. The surface shape of the inward-convex portions **14** is spherical. As described before, the thickness of the cover member **1** is almost uniform all over the whole. The portions corresponding to the inward-convex portions **14** of the cylindrical portion **12** are recessed viewed from the outside. Each inward-convex portion **14** is elastically deformable, similar to the corrugated section **13**.

(41) Further, when the cover member **1** is attached to the thermostat device **10**, the frame **19** is to be moved downward the inward-convex portions **14** by climbing over the inward-convex portions **14**. When the frame **19** is moved downward over the inward-convex portions **14**, the inward-convex portions **14** are caused to elastically deform. After the frame **19** climbs over the inward-convex portions **14**, the inward-convex portions **14** return to the shape before due to their elasticity.

(42) As described above, the cover member **1** of the present embodiment is attached to the thermostat device **10** detachably and, together with the thermostat device **10**, constitutes a thermostat device with the cover member **100**. The thermostat device **10** includes a housing **2** being provided with a body **20** that has an opening **20c** at one end and a groove **20d** formed so as to surround the opening **20c** and a pair of legs **21, 21** extending downward opposing to each other from the lower end opening edge of the body **20** (standing up opposing to each other from the lower end opening edge) inside the groove **20d**, and a gasket **25** to be fitted in the groove **20d**. The cover member **1** includes a dish-shaped portion **11** having insertion holes inside through which the pair of legs **21, 21** are inserted and covering the gasket **25** and a cylindrical portion **12** extending downward from the opening edge of the insertion hole at the center of the dish-shaped portion **11** (standing up from the opening edge of the insertion hole) inside of which the pair of legs **21, 21** are press-fitted. A corrugated section **13** is formed at the portion of the cylindrical portion **12** to which the legs **21** oppose.

(43) With the above constitution, the gasket **25** is covered with the dish-shaped portion **11** in the state where the cover member **1** is attached to the thermostat device **10**, that is, in the thermostat device with the cover member **100**. The coming off of the gasket **25** due to the vibration during transportation, for example, can be surely prevented by attaching the cover member **1** to the thermostat device **10** to form the thermostat device with the cover member **100**.

(44) Since, with the above constitution, the cover member **1** is supported by the thermostat device **10** by press-fitting the legs **21** into the elastically deformable corrugated section **13**, the cover

member **1** is easily detachable to the thermostat device **10**. Further, since providing the corrugated section **13** on the cover member **1** enables it to attach the cover member **1** to the thermostat device **10**, the cover member **1** can be formed by an inexpensive method such as blow molding. Thus, the cover member **1** and the thermostat device with the cover member **100** can be inexpensive.

(45) Furthermore, the forming method and the material of the cover member **1** are appropriately changeable.

(46) In the embodiment of the cover member **1** and the thermostat device with the cover member **100**, the lower end (one end) of the cylindrical portion **12** of the cover member **1** is closed. This allows the prevention of foreign matters such as dust from entering the cover member **1** and, after the thermostat device **10** is assembled to the counterpart member, allows the inhibition of foreign matters from mixing into the coolant. Moreover, the leakage of the coolant due to a bitten foreign matter between the valve body **15** and the valve seat **20b** or between the gasket **25** and the counterpart member can be prevented.

(47) This is particularly effective for the case where the leakage of the coolant needs to be surely inhibited, or the thermostat device **10** is exposed to an environment full of dust; however, an opening may be formed at the lower end (one end) or other portion of the cylindrical portion **12**.

(48) In the present embodiment, the thermostat device **10** includes a valve seat **20b** formed inside the body **20**, a thermo-element **17**, the upper end (one end) of which is supported by the body **20**, extending and retracting depending on the temperature, a valve body **15** unseated from or seated on the valve seat **2b** in response to the extension-retraction operation of the thermo-element **17**, a coil spring (a biasing member) **16** biasing the valve body **15** upward (to the valve seat **20b** side), and a frame **19** being supported by the pair of legs **21, 21** to support the lower end (one end) of the coil spring **16**. Furthermore, an inward-convex portion **14** protruding inward is formed on the cylindrical portion **12**. The inward-convex portion **14** is located at the upper side (body **20** side) than the frame **19** in a state where the cover member **1** is attached to the thermostat device **10**. When the cover member **1** is attached to or detached from the thermostat device **10**, the inward-convex portion **14** elastically deforms due to the contact of the inward-convex portion **14** with the frame **19**.

(49) Since, according to the above constitution, the frame **19** interferes with the inward-convex portion **14** when the cover member **1** is detached from the thermostat device **10**, the cover member **1** is prevented from unintentionally detaching from the thermostat device **10** due to the vibration during the transportation.

(50) The cover member **1** is supported by the thermostat device **10** by hooking the inward-convex portion **14** to the frame **19**, and in such a case, the corrugated section **13** may be eliminated, and the pair of legs **21, 21** may be inserted into the cylindrical portion **12**. In this case, as well, the cover member **1** is easily attached to or detached from the thermostat device **10**. Furthermore, since the cover member **1** is easily attached to the thermostat device **10** by forming the inward-convex portion **14** to the cover member **1**, the cover member **1** can be formed by an inexpensive method such as blow molding. Thus, the cover member **1** and the thermostat device with the cover member **100** can be inexpensive.

(51) The surface shape of the inward-convex portion **14** is spherical in the present embodiment. For this reason, even when the hooking margin of the inward-convex portion **14** and the frame **19** is made large for being hard to come off of the cover member **1** from the thermostat device **10**, it is possible to inhibit the difficulty of attaching the cover member **1** to the thermostat device **10**. That is, it is easy to ensure the hooking margin of the inward-convex portion **14** with the frame **19** and is hard to come off without degrading the mountability of the cover member **1**.

(52) In the present embodiment, the portion of the cylindrical portion **12** located outside of the inward-convex portion **14** is recessed. Since, with this constitution, the thickness of the inward-convex portion **14** can be the same as the other portion, the cover member **1** can be easily formed by blow molding. Moreover, since the inward-convex portion **14** can be easily elastically

deformable, even when the hooking margin of the inward-convex portion **14** and the frame **19** is made large, it can inhibit the difficulty of attaching the cover member **1** to the thermostat device **10**. With this constitution, it can also be hard to come off without degrading the mountability of the cover member **1**.

(53) The shape, wall thickness, number, and position of the inward-convex portion **14** can be appropriately changed. When the unintentional coming off of the cover member **1** from the thermostat device **10** can be sufficiently inhibited by such as setting the fastening margin of the corrugated section **13**, the inward-convex portion **14** can be eliminated.

(54) In the present embodiment, the protrusions **25a**, **25b** protruding outward of the dish-shaped portion **11** are formed on the gasket **25**. With this constitution, the attachment of the gasket **25** can be easily visually confirmed even when the cover member **1** is attached to the thermostat device **10**. This is particularly effective when the cover member **1** is opaque, or the gasket **25** is difficult to visually confirm the gasket **25** from the outside of the cover member **1** even when transparent. However, the protrusions **25a**, **25b** can be omitted if the gasket **25** is surely attached.

(55) The protrusions **25a**, **25b** are used for the purpose of fitting an irregular-shaped (shaped other than true-circular) gasket **25** into the groove **20d** with the correct orientation. In such a case, when the protrusion **25a** protrudes outside the housing **2**, the gasket **25** may come off by catching the protrusion **25a** when handling the thermostat device **10**. Then, since, if the cover member **1** is attached, the gasket **25** does not easily come off if catching the protrusion **25a**, attaching the cover member **1** is particularly effective in the thermostat device **10** with the gasket **25** having the protrusion **25a**.

(56) The shape of the gasket **25** may be true-circular. Then, the protrusions **25a**, **25b** may have a length hidden by the dish-shaped portion **11** or omitted.

(57) In the present embodiment, the bending portions of the mountain fold and the valley fold of the corrugated section **13** are respectively curved with a rounded profile. For this reason, even when the fastening margin of the corrugated section **13** to the legs **21** is made large to prevent the cover member **1** from coming off from the thermostat device **10**, the difficulty in attaching the cover member **1** to the thermostat device **10** is inhibited. That is, even the above constitution allows the prevention of coming off of the cover member **1** without deteriorating the mountability thereof.

(58) The number, shape, and convolution pitch of the bending portions of the mountain fold and the valley fold of the corrugated section **13** may be appropriately changed.

(59) In the present embodiment, the legs **21** are press-fitted into the cylindrical portion **12**, and any other portions except the legs **21** of the thermostat device **10** do not interfere with the cover member **1**. Because of this, the cover member **1** can be easily attached to or detached from the thermostat device **10**.

(60) In contrast, for example, in the cover member **1**, a small diameter portion **12a** may be provided with a corrugated section **13** and a guide portion **26a** of the guide member **26** may be press-fitted thereinside.

(61) In this way, portions of the thermostat device **10** other than the legs **21** may be press-fitted into the cover member **1**.

(62) Preferred embodiments of the invention have been described in detail; modifications, variations, and alterations are possible as long as they do not depart from the scope of the claims.

INDUSTRIAL APPLICABILITY

(63) As described above, the cover member according to the present invention is useful as a temporarily attached cover member to the thermostat device and is particularly suitable for the cover member attached during the transportation of the thermostat device in the term from the manufacturing of a thermostat device as a part to the assembly to a counterpart member.

REFERENCE SIGNS LIST

(64) **1** cover member **2** housing **3** flange **3a** bolt hole **4** flow passage **10** thermostat device **11** dish-shaped portion **12** cylindrical portion **12a** small-diameter portion **12b** intermediate portion **13**

corrugated section **14** inward-convex portion **15** valve body **16** coil spring (biasing member) **17** thermo-element **18** piston **19** frame **20** body **20a** boss **20b** valve seat **20c** opening **20d** groove **21** leg **23** connecting portion **25** gasket **25a** protrusion **25b** protrusion **25c** sealing portion **26** guide member **26a** guiding portion **26b** brim **26a1** lateral hole **30** temperature-sensitive case **100** thermostat device with a cover member

Claims

1. A cover member detachably attached to a thermostat device which is provided with a housing that includes a body having an opening at one end where a groove is formed to surround the opening, and a pair of legs standing up opposingly from one end of the body inside the groove, and with a gasket to be fitted into the groove, comprising: a dish-shaped portion covering the gasket having insertion holes inside in which the pair of legs are inserted; and a cylindrical portion standing up from the opening edge of the insertion holes, inside of which the pair of legs are press-fitted, wherein a corrugated section is formed at a portion opposing to each one of the legs of the cylindrical portion, wherein the thermostat device includes a valve seat formed inside the body, a thermo-element, one end of which is supported by the body, extending and retracting depending on the temperature, a valve body unseated from or seated on the valve seat in response to the extension-retraction operation of the thermo-element, a biasing member biasing the valve body to the valve seat side, and a frame being supported by the pair of legs and supporting one end of the biasing member, and wherein an inward-convex portion protruding inward is formed in the cylindrical portion, the inward-convex portion is located on the body side from the frame in a state where the cover member is attached to the thermostat device, and the inward-convex portion comes into contact with the frame to deform elastically when the cover member is attached to or detached from the thermostat device.
2. The cover member in claim 1, wherein the surface shape of the inward-convex portion is spherical.
3. The cover member of claim 2, wherein portions of the cylindrical portion located at the outside of the inward-convex portion are recessed.
4. A cover member that is detachably attached to a thermostat device, which includes a housing provided with a body that has an opening at one end and a groove formed so as to surround the opening and a pair of legs standing up opposing to each other at the one end of the body inside the groove, a gasket to be fitted in the groove, a valve seat formed inside the body, a thermo-element, one end of which is supported by the body, extending and retracting depending on the temperature, a valve body unseated from or seated on the valve seat in response to the extension-retraction operation of the thermo-element, a biasing member biasing the valve body to the valve seat side, and a frame being supported by the pair of legs and supporting one end of the biasing member, wherein the cover member includes a dish-shaped portion that has insertion holes inside, through which the pair of legs are to be inserted, and covers the gasket, and a cylindrical portion standing up from the opening edge of the insertion holes and to which the pair of legs is to be inserted thereinside, and wherein an inward-convex portion protruding inward is formed in the cylindrical portion, the inward-convex portion is located on the body side from the frame in a state where the cover is attached to the thermostat device, and the inward-convex portion comes into contact with the frame to deform elastically when the cover member is attached to or detached from the thermostat device.
5. The cover member in claim 4, wherein one end of the cylindrical portion is closed.
6. The cover member in claim 4, wherein the surface shape of the inward-convex portion is spherical.
7. The cover member of claim 6, wherein portions of the cylindrical portion located at the outside of the inward-convex portion are recessed.

8. The cover member in claim 4, wherein the gasket is formed with a protrusion protruding outside of the dish-shaped portion.
9. A thermostat device equipped with a cover member, wherein the thermostat device and the cover member are detachably attached to each other, wherein the thermostat device comprises a housing that includes a body having an opening on one end and a groove formed so as to surround the opening and a pair of legs standing up opposing to each other from the one end of the body inside the groove, and a gasket to be fitted in the groove; the cover member is provided with a dish-shaped portion having insertion holes there inside through which the pair of legs are to be inserted, and capped on the gasket; and a cylindrical portion standing up from the opening edge of the insertion holes and inside of which the pair of legs is to be press-fitted; and a corrugated section is formed at a portion opposing to the legs of the cylindrical portion, wherein the thermostat device includes a valve seat formed inside of the body, a thermo-element whose one end is supported by the body extends and retracts depending on the temperature, a valve body that is unseated from or seated on the valve seat in response to the extension-retraction operation of the thermo-element, a biasing member that biases the valve body to the valve seat side, and a frame supported by the pair of legs supporting one end of the biasing member; and wherein an inward-convex portion protruding inward is formed inside the cylindrical portion of the cover member; the inward-convex portion is located at the body side inner than the frame in a state where the cover is attached to the thermostat device, and when the cover member is attached to or detached from the thermostat device, the inward-convex portion comes into contact with the frame to deform elastically.
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